

IEOR 4004 — Quiz 1

Solutions

Exercise 1

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1) Dual

Associate $y_1 \in \mathbb{R}$ (free) with the equality and $y_2 \geq 0$ with the $\leq$ constraint. The dual is

$$\begin{aligned} & \min y_1 - y_2 \\ & \text{s.t. } \begin{cases} y_1 + y_2 = 1 & (\text{column of } x_1, \text{ and } x_1 \text{ is free} \Rightarrow \text{equality}) \\ 2y_1 - y_2 \geq 7 & (\text{column of } x_2, \text{ and } x_2 \geq 0) \\ y_2 \geq 0, y_1 \text{ free.} \end{cases} \end{aligned}$$

2) Complementary slackness

On primal variables:

- $x_2 > 0 \Rightarrow$ the second dual constraint is satisfied at equality.

On dual variables:

- $x_y > 0 \Rightarrow$ the second primal constraint is satisfied at equality.

3) Optimality of $(x_1, x_2) = (-\frac{1}{3}, \frac{2}{3})$

$\bar{x}$ is optimal if and only if it is feasible for the primal and the solution to the following system

- $y_1 + y_2 = 1$ (first dual constraint)
- $2y_1 - y_2 \geq 7$ (second dual constraint, which must be satisfied at equality by complementary slackness)

is a feasible dual solution. We can check that $\bar{x}$ is feasible for the primal. The solution to the system is $(8/3, -5/3)^T$. It violates the nonnegativity constraint on the dual. Hence, it is not optimal for the dual, and $\bar{x}$ is not optimal for the primal.

Exercise 2

1)

$$\begin{aligned}
 \max \quad & \sum_{s \in \{A, B, C\}} (6x_{T,s} + 3x_{S,s} + 8x_{P,s}) \\
 \text{s.t.} \quad & \sum_s x_{T,s} \leq 100, \quad \sum_s x_{S,s} \leq 130, \quad \sum_s x_{P,s} \leq 200 \quad (\text{stock}) \\
 & 2x_{T,A} + 4x_{S,A} + 3x_{P,A} \geq 100 \quad (\text{shop } A \text{ fashionability}) \\
 & 2x_{T,B} + 4x_{S,B} + 3x_{P,B} \geq 120 \quad (\text{shop } B \text{ fashionability}) \\
 & 2x_{T,C} + 4x_{S,C} + 3x_{P,C} \geq 120 \quad (\text{shop } C \text{ fashionability}) \\
 & x_{p,s} \geq 0 \quad \forall p \in \{T, S, P\}, s \in \{A, B, C\}.
 \end{aligned}$$

2)

Demands (units) by shop are defined by the 3×3 matrix d with entries

	A	B	C
T	40	10	20
S	40	60	20
P	40	40	70

Shipping costs per unit to A, B, C : $c_A = 10$, $c_B = 5$, $c_C = 6$, independent of product.

To *satisfy the demand in each shop* (i.e., exactly meet it), impose

$$x_{p,A} = d_{p,A}, \quad x_{p,B} = d_{p,B}, \quad x_{p,C} = d_{p,C} \quad \forall p \in \{T, S, P\}.$$

(Equivalently, use “greater or equal” constraints).

Letting π_p be the profit for product p , the objective function becomes to maximize profit minus shipping costs:

$$\max \sum_s \sum_p (\pi_p - c_s) x_{p,s}.$$

With the equalities above, the LP reduces to checking stock feasibility (already satisfied), but for a uniform formulation one can keep stock constraints as $\sum_s x_{p,s} \leq U_p$ (redundant under the given demands).

3)

Let the realized fashionability at shop s be

$$F_s := 2x_{T,s} + 4x_{S,s} + 3x_{P,s}.$$

Require each shop to be at least 80% of every other shop:

$$F_s \geq 0.8 F_t \quad \forall s, t \in \{A, B, C\}.$$

Bonus

By definition,

$$C_1 = \{\hat{c}_1 : x^* \text{ is optimal when the cost of } x_1 \text{ is } \hat{c}_1 \text{ and the cost of } x_2 \text{ is } c_2\}.$$

Thus, in particular, for every value $\hat{c}_1 \in C_1$ we have that $(\hat{c}_1, c_2) \in C$. We conclude that the correct answer is b). An example where there are values $(\hat{c}_1, \hat{c}_2) \in C$ for which however $\hat{c}_1 \notin C_1$ is the following.

$$\max x_1 + x_2 : x_1 + 2x_2 \leq 2; 2x_1 + x_2 \leq 2; x \geq 0,$$

with optimal solution $x^* = (2/3, 2/3)^T$. Here, $.1 \notin C_1$ but $(.1, .1) \in C$. This shows that a) is, in general, false.